SAVEETHA

INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Course Code: **CSA51**

Course Name: **Cryptography and Network Security**

Duration: **1: 30 Hr**

Date : Max Marks: **50 Marks**

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for ensuring data security.
CO2	Analyze public-key cryptosystem strategies using RSA and key exchange for strengthening information security.

Answer all the Questions

1. A security analyst at a small firm is asked to demonstrate classical encryption to new trainees. The plaintext "MEETATNOON" must be encrypted using a Caesar cipher with shift 7. Show the encryption and decryption process step by step, first converting each letter to its numeric value (A=0, B=1, …, Z=25), applying the formulas C = (P + 7) mod 26 and P = (C − 7) mod 26, and then converting the numbers back to letters. Next, encrypt the same plaintext with a Vigenère cipher using the repeating key "SUN", showing the key shift for every letter position. Compare the Caesar cipher with the Vigenère cipher in terms of

   a) key space

   b) treatment of repeated patterns such as the double letters EE and OO and the repeated letters T and N, and

   c) vulnerability to cryptanalysis

   Recommend one of the two only for a constrained educational demonstration, and justify

your choice. Finally, explain why neither cipher is suitable for protecting modern confidential data, considering key size, determinism, lack of integrity protection and the strength of modern algorithms such as AES-256-GCM. **(10) (CO1) (BL3)**

2. A cryptography instructor wants to show how matrix algebra is used in classical block ciphers.

Encrypt the plaintext "MATH" using a 2 × 2 Hill cipher with key K = [[3, 2], [5, 7]], using A=0, B=1,..., Z=25. Show the conversion of the plaintext into column vectors (MA and TH), and show the matrix multiplication and modulo-26 reduction for each block to obtain the ciphertext.

Determine whether the key matrix is invertible modulo 26 by computing its determinant and checking gcd(det, 26) = 1. Find the modular inverse of the determinant, compute the inverse key matrix K^{-1} using the adjugate method, and verify that $K \cdot K^{-1} \equiv I \pmod{26}$. Explain how K^{-1} enables decryption by applying it to the ciphertext blocks and recovering "MATH". Show why the alternative key K = [[2, 3], [4, 5]] is unacceptable by calculating its determinant and its gcd with 26. Finally, identify and explain a major weakness of the Hill cipher, such as its linearity and vulnerability to known-plaintext attacks. **(10) (CO1) (BL4)**

3. a) A hospital network of 200 clinics must send large MRI archives (about 1 GB each) to a central radiology server, and every clinic must also exchange encryption keys with the server securely. The IT team is choosing between a symmetric cipher (AES-256-GCM) and the RSA public-key algorithm. Compare the two in terms of

a) key usage (who holds which key),
b) performance and maximum data size per operation,
c) scalability, including the number of keys needed if all 200 clinics must communicate pairwise versus using RSA key pairs, and key-distribution requirements, including the need for certificates to authenticate public keys. (5) (CO1) (BL4)

b) Using this comparison, design a hybrid encryption workflow that uses each algorithm where it is most appropriate, listing the steps carried out by the sending clinic (certificate validation, DEK and nonce generation, AES-GCM encryption with associated data, RSA OAEP key wrapping, and packaging) and by the receiving server (key unwrapping, decryption, and tag verification). Also explain how secure key storage should be handled (HSM/KMS, non-exportable keys, role-based access, MFA, audit logging, backup and recovery), and how key rotation should be handled (per-file DEKs, periodic rotation of the key-encryption key, re-wrapping, retirement of old keys, and emergency rotation after suspected compromise). **(5) (CO2) (BL4)**

4. For an RSA demonstration in a cryptography laboratory, let the primes be p = 11 and q = 19, with public exponent e = 7. Calculate the modulus n and Euler's totient $\varphi(n)$, and confirm that $\gcd(e, \varphi(n)) = 1$. Determine the private exponent d such that $e \cdot d \equiv 1 \pmod{\varphi(n)}$ using the extended Euclidean algorithm, showing every division step and the back substitution. State the public key and the private key. Encrypt the message M = 5 using C = M^e mod n, showing the intermediate modular exponentiation steps. Verify the result by decrypting the ciphertext using M = C^d mod n (you may use the Chinese Remainder Theorem to simplify the calculation) and confirm that the original message is recovered. Finally, explain why real systems require much larger parameters (2048-bit or 3072-bit moduli) and careful padding such as RSA-OAEP for encryption and RSA-PSS for signatures, discussing factoring attacks, determinism, malleability and small-message attacks on textbook RSA. **(10) (CO2) (BL4)**

5. A financial institution transmits confidential customer information across a network and

must protect it against unauthorized access. Apply suitable classical encryption techniques, such as Caesar cipher, Playfair cipher, and Vigenère cipher, to encrypt and decrypt the given plaintext. Extend your solution by applying an appropriate symmetric or asymmetric cipher to secure the same information. Demonstrate the encryption and decryption procedures, compare the techniques in terms of key usage and security, and justify which approach is more suitable for protecting sensitive financial data. **(10) (CO1) (BL3)**

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CIA - I

1) Caesar Cipher and vigenere Cipher

Given:

-) plaintext = MEETATNOON
-) Caesar shift = 7
-) Vigenere key = SUN
-) A=0, B=1 .... Z=25.

A) Caesar Cipher Encryption:-

formula:-

$$C = (P+7) \bmod 26.$$

plaintext	M	E	E	T	A	T	N	O	O	N
P	12	4	4	19	0	19	13	14	14	13
P+7	19	11	11	26	7	26	20	21	21	20
.S	19	11	11	0	7	6	20	21	21	20
Ciphertext	T	L	L	A	H	A	U	V	V	U

Encrypted Text:

B) Caesar Cipher Decryption:-

formular

$$P = (C-7) \bmod 26.$$

Ciphertext	T	L	L	A	H	A	U	V	V	U
C	19	11	11	0	7	0	20	21	21	20
C-7 mod 26	12	4	4	19	0	19	13	14	14	13
plaintext	M	E	E	T	A	T	N	O	O	N

Decrypted Text:= MEETATNOON.

C) Vigenere Cipher Encryption:-

Given key=SUN

Repeating the key to match plaintext

Plaintext : MEET AT NOON

key : SUNSUN SUNS

formula:

$$e = (P + k) \bmod 26.$$

plaintext	M	E	E	T	A	T	N	O	O	N	
P	12	4	4	19	0	19	13	14	14	13	
key	S	U	N	S	U	N	S	U	N	S	
k	18	20	13	18	20	13	18	20	13	18	
C	4	24	17	11	20	6	5	8	1	5	
Ciphertext	E	Y	R	L	U	G	F	I	B	F.	

Encrypted Text = "EYRLUGFIBF"

D) Vigenere Decryption:

formula:

$$P = (C - k) \bmod 26.$$

subtract the corresponding key values.

Ciphertext	E	Y	R	L	U	G	F	I	B	F	
C	4	24	17	11	20	6	5	8	1	5	
k	18	20	13	18	20	13	18	20	13	18	
P	12	4	4	19	0	19	13	14	14	13	
Plaintext	M	E	E	T	A	T	N	O	O	N	

Decrypted Text = "MEETATNOON".

E.) Comparison:-

Criteria	Caesar Cipher	Vigenere Cipher
Key Space	26 possible shifts	$26^3 = 17576$ for a three-letter key
Repeated patterns	preserves Repeated letters	Can encrypt repeated letters differently depending on key position.
Cryptanalysis	Easily broken using brute force and frequency analysis	More resistant, but vulnerable to Kasiski and frequency analysis.

F). Recommendation and Modern Security:

Recommendation :- Caesar Cipher is suitable for a constrained educational demonstration because it is simple and helps beginners understand substitution and modular arithmetic

Neither Cipher is suitable for modern confidential data because :

→ They have very small key spaces

→ They are deterministic and reveal repeated patterned patterns

→ They don't provide integrity or authentication

→ They are vulnerable to cryptanalysis.

AES-256-GCM is preferred because it provides strong encryption, confidentiality, integrity, and authentication using a 256-bit key.

2) Hill Cipher Encryption and Decryption :-

Given:-

Plaintext: MATH

key Matrix

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

A=0, B=1, ... Z=25.

A) Plaintext Conversion:

Divide plaintext into two blocks: MA and TH

$$MA = \begin{bmatrix} 12 \\ 0 \end{bmatrix} \quad TH = \begin{bmatrix} 19 \\ 7 \end{bmatrix}$$

B) Encryption:

Formula:

$C = KP \bmod 26$

Block 1: MA

$$C_1 = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 36 \\ 60 \end{bmatrix}$$

Modulo 26:

$$= \begin{bmatrix} 36 \bmod 26 \\ 60 \bmod 26 \end{bmatrix} = \begin{bmatrix} 10 \\ 8 \end{bmatrix}$$

10=K, 8=I.

Block 2: TH

$$C_2 = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 19 \\ 7 \end{bmatrix} = \begin{bmatrix} 57+14 \\ 95+49 \end{bmatrix} = \begin{bmatrix} 71 \\ 144 \end{bmatrix}$$

Modulo 26:

$$= \begin{bmatrix} 71 \bmod 26 \\ 144 \bmod 26 \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix}$$

19=T, 14=O

Ciphertext:- KITO.

C) Checking key Invertibility.

For the key Matrix:

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

Determinant: $|K| = (3 \times 7) - (2 \times 5) = 21 - 10 = 11.$

Check:

$$\gcd(11, 26) = 1$$

Therefore, the key Matrix is invertible modulo 26.

D) Modular Inverse of Determinant:

Find $11^{-1} \bmod 26$.

$$11 \times 19 = 209.$$

$$209 \bmod 26 = 1$$

$$\therefore 11^{-1} = 19$$

E) Finding Inverse key Matrix:

Adjugate method:-

$$K^{-1} = 11^{-1} \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$$

Substitute:

$$= 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \bmod 26.$$

$$\boxed{K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}}$$

F) verification:

$$KK^{-1} = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} = \begin{bmatrix} 27 & 52 \\ 78 & 105 \end{bmatrix}$$

Modulo 26:

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Hence,

$$KK^{-1} \equiv I \pmod{26}$$

(7) Decryption formula:

$$P = K^{-1} C \bmod 26.$$

Block 1: k1

$$P1 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 10 \\ 8 \end{bmatrix} = \begin{bmatrix} 30+112 \\ 90+40 \end{bmatrix} = \begin{bmatrix} 14 \\ 130 \end{bmatrix}$$

Modulo 26:

$$= \begin{bmatrix} 12 \\ 0 \end{bmatrix} \rightarrow MA.$$

Block 2: TO

$$P2 = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 19 \\ 14 \end{bmatrix} = \begin{bmatrix} 57+196 \\ 171+70 \end{bmatrix} = \begin{bmatrix} 253 \\ 241 \end{bmatrix}$$

Modulo 26:

$$= \begin{bmatrix} 19 \\ 7 \end{bmatrix} \rightarrow TH.$$

Recovered plaintext: MATH.

4) Alternative key verification:-

Given:

$$K = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$$

Determinant:

$$|K| = (2 \times 5) - (3 \times 4)$$
$$= 10 - 12 = -2$$
$$\gcd(-2, 26) = 2$$

Since GCD is not 1, the matrix is not invertible modulo 26

∴ this key is unacceptable

I) Major weakness:-

The Hill Cipher is linear, making it vulnerable to known plaintexts attacks. If attackers obtain sufficient plaintext-ciphertext pairs, they can determine the encryption key matrix and recover the original messages.

3) AES -256 -GCM and RSA - Comparison:-

3(a). Comparison of AES and RSA:

Given:

a) Number of clinics = 200
b) MRI file size = 1GB
c) Algorithms = AES-256-GCM and RSA.

Comparison Table:

Criteria	AES-256-GCM	RSA
Key usage	Same secret key for encryption and decryption	public key encrypts, private key decrypts
Performance	Very fast	comparatively slow
Data Size	Supports large files	Limited plaintext size per operation
Key distribution	Secret key must be securely shared	public key can be distributed using certificates
Scalability	Requires secure pairwise key management	Each entity maintains a public-private key pair.

Number of keys Required:-

1) AES pairwise communication.

Formula:- $\frac{n(n-1)}{2} = \frac{200(199)}{2} = 19900$ keys.

2) RSA pairs:-

Each clinic requires one key pair and the central server requires one key pair.

$200+1 = 201$ key pairs

Each clinic uses the server's authenticated public key to encrypt a data-encryption key. RSA public keys should be authenticated using digital certificates issued by a trusted Certificate Authority (CA).

Conclusion:-

AES-256-GCM is suitable for large MRI archives because of its speed, whereas RSA is suitable for securely exchanging encryption keys.

3(b) Hybrid Encryption Workflow:-

A Hybrid encryption System combines AES and RSA to provide efficient and secure communication.

A) Sending clinic:-

1) Certificate validation: Validate the central Server's RSA public key certificate.

2) DEK Generation: Generate a random 256-bit Data Encryption key (DEK)

3) Nonce Generation: Generate a unique 96-bit nonce for AES-GCM.

4) AES Encryption:- Encrypt the ICTB MRI file using AES-256-GCM.

5) RSA key Wrapping:- Encrypt the DEK using the server's RSA public key with RSA-OAEP padding.

6) packaging: Send the encrypted MRI file, encrypted DEK, nonce, authentication tag and associated metadata.

B) Receiving Server:-

1) Receive the encrypted package.

2) Use the RSA private key to unwrap the DEK using RSA-OAEP.

3) Use the recovered DEK and nonce to decrypt the MRI file using AES-256-GCM.

4) verify the authentication tag and associated data.

5) If verification succeeds, accept the original MRI file; otherwise reject the file.

C) Secure key Storage:-

- HSM / KMS:- Store keys in hardware Security modules or key Management Systems.
  -) Non exportable keys: Prevent private keys from being directly extracted.
  -) Role based Access Control:- Allow only authorized persons to access keys.
  -) MFA: Require multi-factor authentication
  -) Audit logging: Record all key access and modification activities.
  -) Backup and Recovery:- Maintain secure backups for disaster recovery.

D) key Rotation:-

1) Generate a new DEK for every file
2) Periodically rotate the RSA key-encryption key.
3) Re-wrap existing DEKs using the new RSA public key where necessary.
4) Retire old keys securely after migration.
5) Perform emergency rotation immediately if Compromise is suspected.

Result:- Hybrid encryption provides efficient bulk data encryption using AES and secure key exchange using RSA.

4) RSA Algorithm - Encryption and Decryption

Given:-

$p=11, q=19, e=7, M=5.$

A) Calculate Modulus:-

formula:-

$n=p\times q \Rightarrow n=11\times 19 \Rightarrow \boxed{n=209.}$ //

B) Calculate Euler's Totient.

$$\phi(n) = (p-1)(q-1)$$
$$= (11-1)(19-1) = 10\times 18 \Rightarrow \boxed{\phi(n)=180.}$$ //

C) verify Public Exponent:..

$$\gcd(e, \phi(n)) = \gcd(7, 180)$$

$\boxed{\gcd(7,180)=1.}$ //

Therefore, the public exponent is valid.

D) Find Privatekey Using Extended Eudicleam Algorithm

We need to find out:..

$$7d \equiv 1 \pmod{180}.$$

Division steps:-

$$180 = 25(7)+5$$
$$7 = 1(5)+2$$
$$5 = 2(2)+1.$$

Back substitution:-

$$1 = 5-2(2) \Rightarrow 5-2(7-5)$$
$$= 3(5)-2(7)$$
$$\Rightarrow 3(180-25(7))-2(7).$$

Otherwise repeat the link.

$= 3(180) - 77(7) \Rightarrow 1 = 3(180) - 77(7)$

Therefore,

$d = -77 \bmod 180 \Rightarrow \boxed{d = 103.}$ //

verification:-

$7 \times 103 = 721 \Rightarrow 721 \bmod 180 = 1.$

E) public and private keys:..

publickey :- $\boxed{(e, n) = (7, 209)}$. //

Privatekey :- $\boxed{(d, n) = (103, 209).}$ //

F) RSA Encryption:-

formula:-

$$C = M^e \bmod n$$

substitute:-

$$C = 5^7 \bmod 209$$

Calculate powers:..

$5^2 = 25$

$5^4 = 625 \bmod 209 = 207$

$5^7 = 5^4 \times 5^2 \times 5$

$\Rightarrow 207 \times 25 \times 5 \bmod 209.$

$\Rightarrow 207 \times 25 = 5175$

$5175 \bmod 209 = 159$

$159 \times 5 = 795 \Rightarrow 795 \bmod 209 = 168.$

Ciphertext $C = 168.$

G) RSA Decryption using CRT:-

formula:-

$$M = C^d \bmod n.$$

$M = 168^{103} \bmod 209.$

Using Chinese Remainder Theorem:-

Step -1:- Modulo $p=11$

$$168 \bmod 11 = 3$$

$$d \bmod (p-1) = 103 \bmod 10 = 3$$

$$M_1 = 3^3 \bmod 11$$

$$= 27 \bmod 11 = \boxed{5}$$

Step-2:- Modulo $q=19$.

$$168 \bmod 19 = 16 = -3$$

$$d \bmod (q-1) = 103 \bmod 18 = 13$$

$$M_2 = (-3)^{13} \bmod 19.$$

$$= -3^{13} \bmod 19.$$

Using modular powers:-

$$3^4 \bmod 19 = 5$$

$$3^8 \bmod 19 = 6$$

$$\Rightarrow 3^{13} = 3^8 \times 3^4 \times 3 = 6 \times 5 \times 3 = 90$$

Therefore, $90 \bmod 19 = 14$.

$$(-3)^{13} \bmod 19 = -14 \bmod 19.$$

$$\boxed{M_2 = 5.}$$

Step-3:- Combine results:-

$$M \equiv 5 \pmod{11}$$

$$M \equiv 5 \pmod{19}$$

Therefore, $\boxed{M = 5.}$

4) Why Real Systems Require Stronger RSA :-

1) Large key sizes : 2048-bit or 3072-bit modulo make integer factorization computationally difficult.

2) Factoring attacks : Small RSA moduli can be broken by factoring n into p and q.

3) Determinism :- Textbook RSA produces identical ciphertexts for identical messages.

4) Malleability :- Attackers can manipulate ciphertext mathematically.

5) Small-message attacks : Small messages may be vulnerable without suitable padding.

Secure padding :-

-) RSA-OAEP is used for encryption.
-) RSA-PSS is used for digital signatures.

5) Classical and Modern encryption Techniques :-

A) Caesar Cipher :-

Given :-

-) Plaintext = HELLO
-) Shift = 3.

Formula :-

$$C = (P+3) \bmod 26.$$

plaintext	H	E	L	L	O
P	7	4	11	11	14
C	10	7	14	14	17
Ciphertext	K	H	O	O	R.

Encrypted Text = KHOOR.

Decryption:-

$$P = (C-3) \bmod 26.$$

KHOOR → HELLO.

B) Playfair Cipher:-

Given:-

- Plaintext = HELLO
- Key = MONARCHY
- I/J combined.

Key Matrix:-

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z.

Divide plaintext into pairs. Insert X b/w repeated letters.

HELLO → HELXLO. → HE I LX I LO.

Encryption rules:-

1) Same row: take the letter to the right.
2) Same Column: take the letter below.
3) Rectangle: take the opposite corner in the same row.

Encryption:-

·) HE → CF
·) LX → SU
·) LO → PM.

·) Ciphertext: CFSUPM.

→ Decryption using the reverse rules

CF | SU | PM.

HE | LX | LO.

→ Remove the filter X

→ Recovered plaintext = HELLO.

C. Vignere Cipher:-

Given:-

·) plaintext = HELLO
·) key = KEY.

Repeat key:-

plaintext: HELLO
key: KEYKE

Formula:-

$$C = (P + K) \bmod 26.$$

Plaintext	H	E	L	L	O
P	7	4	11	11	14
key	K	E	Y	K	E
K	10	4	24	10	4

C	17	8	9	21	18
Ciphertext	R	I	J	V	S.

Encrypted Text = RIJVS.

Decryption :-

$$P = (C - K) \bmod 26.$$

RIJVS → HELLO.

D) Modern Encryption - AES-256-GCM :-

AES-256-GCM is a modern symmetric authenticated encryption algorithm.

Encryption Procedure :-

1) Generate a random 256-bit secret key.
2) Generate a unique 96-bit nonce.
3) Apply AES-256 GCM to plaintext HELLO.
4) Generate Ciphertext and authentication tag.
5) Transmit the Ciphertext, nonce and authentication tag securely.

Decryption procedure :-

1) Receive the encrypted data.
2) Use the same secret key and nonce.
3) Decrypt the Ciphertext.
4) Verify the authentication tag.
5) Recover HELLO if verification succeeds.

e) Comparison of Encryption Techniques:-

Technique	key usage	Security.
Caesar	Single numerical shift	Very weak
playfair	5×5 key matrix	weak.
Vigenere	Repeating keyword	Moderate historically; weak today.
AES-256-GCM	256-bit secret key	Strong modern authenticated encryption.

f) Recommendation:-

AES-256-GCM is the most suitable technique for protecting sensitive financial information because:-

1) It provides strong confidentiality.
2) It supports large amounts of financial data.
3) It provides integrity and authentication.
4) It detects unauthorised modification
5) It is widely used in secure communication systems.

Conclusion:-

Classical ciphers are useful for understanding encryption principles., but AES-256-GCM is recommended for securing confidential financial information.